MIL-F-14716A (MU)
22 MARCH 1971
SUPERSEDING
MIL-F-14716 (MU)
30 June 1968

MILITARY SPECIFICATION

FUZE, TIME, M84A1
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of time fuze designated as the M84A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

PURCHASE DESCRIPTIONS

PA-PD-3056	-	Tungsten Powder (For Pyrotechnics)
PA-PD-3091	-	Delay, Tungsten, Composition

MILITARY

MIL-P-223	-	Powder, Black
MIL-A-2550	-	Ammunition and Special Weapons; General Specification for
MIL-P-12951	-	Primer, Percussion, M39A1; Loading, Assembling and Packing
MIL-I-45607	-	Inspection Equipment; Supply and Maintenance of
MIL-P-46994	-	Pellets - Boron Potassium Nitrate

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes; (ABC-STD-105).
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-331	-	Jumble Test for Use in Production of Fuzes

FSC: 1390

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1234 - Pyrotechnics; Sampling, Inspection and Testing
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

9232784 - Fuze, Time, M84A1
8865235 - Container, Ammunition, Fiber, M200A1, for M84A1 Time Fuze
8865236 - Box, Packing, Ammunition, for M84A1 Time Fuze in Fiber Container, M200A1

PUBLICATIONS

ARMY

ET-9232784 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Fuze assembly.-The fuze assembly shall comply with all requirements specified on Drawing (dwg.) 9232784 and with all requirements specified in applicable specifications.

3.3 Moisture content

3.3.1 Black powder.-The moisture content of the black powder, at the loading station at the time of loading the charges in the fuze assembly, shall not exceed 0.7 percent when tested as specified in 4.3.1.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

3.4 Volatiles content

3.4.1. Delay composition.-The volatiles content of the delay composition at the loading station at the time of loading the charges in the fuze assembly, shall not exceed 0.1 percent when tested as specified in 4.3.2.2.

3.4.2 Boron potassium nitrate pellets.-The volatile content of the boron potassium nitrate pellets, at the loading station at the time of loading the charges in the fuze assemblies, shall not exceed 0.75 when tested as specified in 4.3.2.1.

3.5 Jumble.-The fuze assembly shall comply with the requirements of Standard MIL-STD-331 when tested as specified in 4.3.4.

3.6 Static functioning and uniformity of burning time.-The fuze assemblies shall function and the lot shall meet the burning time criteria shown in Table I when tested in accordance with 4.2.3.4.

TABLE I

Fuze Setting (sec)	Burning Time (sec)
20	17.85 - 21.15
30	27.25 - 30.75

3.7 Functioning (proving ground).-The fuze assembly shall comply with the following requirements when conditioned for 4 hours at 75° plus or minus (±) 5°F, and tested as specified in 4.3.5. (The requirements of 3.7, 4.2.1.1.1(e), 4.2.1.1.3, 4.2.1.1.4, 4.2.3.6, including Table III and 4.3.5 are not applicable when the contractor for the fuze LAP is also the contractor for the assembling and packing of the complete round).

3.7.1 Dud.-The fuze assembly shall not fail to function.

3.7.2 Premature burst.-There shall be no premature burst or metal parts failure in the barrel of the mortar, or within 200 yards horizontal range of the mortar.

3.7.3 Ballistic functioning and burning time.-The fuze assemblies during ballistic testing shall function and shall comply with the burning time indicated in Table II when tested in accordance with 4.3.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

TABLE II

Fuze Setting (sec)	Charge Increment	Quadrant Elevation (Degrees)	Burning Time (sec)
20	4	60	18.10 - 21.90
30	8	60	28.00 - 32.00

3.8 Workmanship.-All parts shall be loaded and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, surface defects, chips, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Percussion primers of one lot interfix number from one manufacturer.

b. Black powder for expelling charge of one lot interfix number from one manufacturer.

c. Delay composition of one lot from one manufacturer.

d. Metal parts of one lot interfix number from one manufacturer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

4.2.1.1 First article inspection

4.2.1.1.1 Submission of fuze assemblies and subassemblies.-The contractor shall submit a first article sample for determination of conformance to drawings and specification requirements. The first article sample shall consist of the following:

a. Thirty body assemblies (dwg. 9232789) complete with assembled components.

b. Fifteen loaded upper time train ring assemblies (dwg. 9232792).

c. Fifteen loaded graduated time train ring assemblies (dwg. 9232791).

d. Thirty fuze assemblies (dwg. 9232784). These fuzes will be tested statically in accordance with 4.2.1.1.2.

e. Thirty fuze assemblies (dwg. 9232784). These fuzes will be tested ballistically in accordance with 4.2.1.1.3.

The above assemblies shall have been produced by the contractor, or furnished by a supplier, using the same production processes, procedures and equipment as will be used in fulfilling the contract. The first article submission (items a through e) will be tested and inspected by the Government for conformance to any or all drawing and specification requirements (see 6.5). Prior to submission, the contractor shall inspect the samples to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the samples, including certificates of analysis for materials and statements of findings for all inspections and tests performed (see 6.6). A sample will not be submitted containing known defects unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever a change occurs in manufacturing process, material used, drawing or specification as to significantly affect product uniformity as determined by the agency responsible for technical data.

4.2.1.1.2 First article static testing.-Thirty fuze assemblies shall be tested as specified in 4.3.3 using equipment specified in 4.2.4. Fifteen samples shall be fired at 20 second setting and fifteen samples shall be fired at 30 second setting. Observations shall be made for the requirements specified in paragraph 3.6 and Table I..

4.2.1.1.3 First article ballistic testing.-The thirty fuze assemblies for ballistic testing shall be tested as specified in paragraph 4.3.5. Fifteen samples shall be fired at 20 second setting and fifteen samples at 30 second setting. Observation shall be made for the requirements specified in paragraph 3.7.3 and Table II. First article ballistic testing will not be performed until a first article static test sample is accepted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

4.2.1.1.4 Recovery.-The proving ground shall make the necessary arrangements to recover any or all malfunctioning fuze assemblies which may occur during the first article ballistic test. Arrangements for the recovery of malfunctions shall be for the first article ballistic test only, unless directed by the procuring activity.

4.2.1.1.5 Rejection.-If any fuze assembly or subassembly fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a complete first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance 4.2.4.

4.2.2.1 Ring, graduated time train, prior to loading (See dwg. 9232791, covering a detail of dwg. 9232784).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Charge groove not coated.....	Visual	01001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8).....	Visual	01002

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

4.2.2.2 Ring, graduated time train, prior to loading pellets and prior to the assembly of non-metallic groove washer. (See dwg. 9232791, covering a detail of dwg. 9232784).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Location of pellet "B" cavity	Gage	02001
102.	Location of ignition vent cavity	Gage	02002
103.	Groove charge missing, cracked, chipped or not full	Visual	02003
104.	Ignition vent cavity not visible through pellet "B" cavity	Visual	02004
105.	Outside diameter of undercut	Gage	02005
106.	Depth of circumferential undercut	Gage	02006
107.	Incorrect number of delay vent cavities	Visual	02007
Minor:	AQL 4.0 percent		
201.	Thickness, minimum (min.)	Gage	02008
202.	Diameter of delay vent cavities	Gage	02009
203.	Depth of delay vent cavities	Gage	02010
204.	Location of delay vent cavities	Gage	02011
205.	Safety disc missing or incorrectly positioned	Visual	02012
206.	Diameter of pellet "B" cavity	Gage	02013
207.	Diameter of ignition vent cavity	Gage	02014
208.	Depth of pellet "B" cavity	Gage	02015
209.	Depth of ignition vent cavity	Gage	02016
210.	Evidence of poor workmanship, (see 3.8)	Visual	02017

4.2.2.3 Ring, graduated time train, prior to assembly of brass disc. (See dwg. 9232791, covering a detail of dwg. 9232784).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Pellet "C" missing, incorrectly positioned or cracked	Visual	03001
Minor:	None defined.		

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

4.2.2.4 Ring, graduated time train assembly. (See dwg. 9232791, covering a detail of dwg. 9232784).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Pellet "B" missing, incorrectly positioned or cracked	Visual	04001
102.	Washer "A" missing, incorrectly positioned or not secure	Visual	04002
103.	Delay vent discs missing, punctured or torn	Visual	04003
104.	Delay vent washers missing or incorrectly assembled	Visual	04004
Minor:	AQL 1.5 percent		
201.	Washer "B" groove cover missing, not secure or damaged to the extent that the groove charge is exposed	Visual	04006
202.	Wax coating incomplete or missing, except that pin holes and bubbles may be ignored	Visual	04007
203.	Evidence of poor workmanship (see 3.8)	Visual	04008

4.2.2.5 Upper time train ring, prior to loading. (See dwg. 9232792, covering a detail of dwg. 9232784).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Charge groove not coated	Visual	05001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	05002

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

4.2.2.6 Upper time train ring, prior to loading pellet, and prior to assembly of non-metallic groove washer. (See dwg. 9232792, covering a detail of dwg. 9232784).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Location of "A" pellet cavity	Gage	06001
102.	Location of ignition vent cavity	Gage	06002
103.	Groove charge missing, cracked, chipped or not full	Visual	06003
104.	Pellet and groove charge cavity not visible through ignition vent cavity	Visual	06004
105.	Incorrect number of delay vent cavities	Visual	06005
106.	O.D. of circumferential undercut	Gage	06006
107.	Depth of circumferential undercut	Gage	06007
Minor:	AQL 4.0 percent		
201.	Thickness, minimum (min.)	Gage	06008
202.	Diameter of delay vent cavities, min.	Gage	06009
203.	Depth of vent cavities	Gage	06010
204.	Location of delay vent cavities	Gage	06011
205.	Diameter of "A" pellet cavity	Gage	06012
206.	Diameter of ignition vent cavity	Gage	06013
207.	Depth of "A" pellet cavity	Gage	06014
208.	Depth of ignition vent cavity	Gage	06015
209.	Evidence of poor workmanship, (see 3.8)	Visual	06016

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

4.2.2.7 Upper time train ring assembly. (See dwg. 9232792, covering a detail of dwg. 9232784).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Pellet "A" incorrect, missing or cracked	Visual	07001
102.	Delay vent discs missing, punctured or torn	Visual	07002
103.	Delay vent washers missing or incorrectly assembled	Visual	07003
Minor:	AQL 1.5 percent		
201.	Washer "B" missing, not secure or damaged to the extent that the groove charge is exposed	Visual	07005
202.	Wax coating incomplete or missing, except that pin holes and bubbles may be ignored	Visual	07006
203.	Evidence of poor workmanship (see 3.8)	Visual	07007

4.2.2.8 Body assembly, prior to securing retainer cover (See dwg. 9232789, covering a detail of dwg. 9232784).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Charge cavity liner missing	Visual	08001
102.	Non-metallic cover on charge cavity liner missing	Visual	08002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	08003

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

4.2.2.9 Body assembly (See dwg. 9232789, covering a detail of dwg. 9232784).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Pellet incorrect, cracked or missing	Visual	09001
102.	Washer missing, not secure or incorrectly assembled	Visual	09002
103.	Retainer cover or disc missing	Visual	09003
104.	Primer missing	Visual	09004
105.	Retainer ring not properly seated or staked	Visual-Manual	09005
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	09006

4.2.2.10 Fuze assembly, prior to assembling head assembly. (See dwg. 9232784).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Component(s) inverted or missing	Visual	10001
Minor:	AQL .065percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	10002

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

4.2.2.11 Fuze assembly (See dwg. 9232784).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Fuze not set safe (within 1/4 inch of safety witness mark)	Visual	11001
2.	Safety wire missing	Visual	11002
Major:	AQL 0.40 percent		
101.	Head assembly loose	Manual	11003
Minor:	AQL 1.5 percent		
201.	Total length, max.	Gage	11004
202.	External threads damaged or contaminated	Visual	11005
203.	Marking missing, incorrect or illegible	Visual	11006
204.	Evidence of poor workmanship (see 3.8)	Visual	11007

4.2.2.12 Sealed fiber Box (inner) (See dwg. 8865235)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 1.0 percent		
101.	Tape not in contact with container flaps	Visual	12001
102.	Container cut or damaged through all impregnated layers	Visual	12002
103.	Metal end loose or distorted	Visual-Manual	12003
104.	Dessicant missing	Visual	12004
105.	Outer bag improperly sealed or damaged	Visual	12005
Minor:	AQL 1.5 percent		
201.	Contents loose	Manual	12006
202.	Marking missing, incorrect or illegible	Visual	12007
203.	Cuts, scuffs, or gouges in outer layers	Visual	12008
204.	Evidence of poor workmanship, (see 3.8)	Visual	12009

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

4.2.2.13 Sealed Fiber box (See dwg. 8865236).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Box damaged to the extent that contents are exposed or are liable to become exposed.....	Visual	13001
102.	Tape missing, broken, improperly sealed or loose.....	Visual Manual	13002
Minor:	AQL 1.0 percent		
201.	Contents loose.....	Visual-Manual	13003
202.	Marking missing, incorrect or illegible.....	Visual	13004

4.2.3 Testing

4.2.3.1 Moisture contents of black powder (see 3.3.1), Code No. 14001, Major defect.-The contractor shall provide adequate controls to insure that all black powder used in the fuze assembly comply with the moisture content requirements. For verification the contractor shall select one sample of each sub-lot of powder used in each eight hours loading production of fuze assemblies. A composite sample shall not be used. If the moisture content of a sample exceeds the requirement and loading has not begun, that quantity of material represented by the sample shall be rejected. If fuze assemblies have been loaded with material containing excessive moisture, the sample, together with all assemblies, shall be rejected. The test shall be performed as specified in 4.3.1.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

4.2.3.2 Volatile content of boron potassium nitrate pellets (see dwg. 9234458) (see 3.4.2), Major defect, Code No. 15001.-The contractor shall provide adequate control to insure that the manufacture of pellets complies with the applicable requirements. Pellets manufactured from acceptable boron potassium nitrate composition shall, prior to assembly in fuzes, be stored under conditions which preclude contamination by volatile and foreign material. For verification, and just prior to the fabrication of pellets, the contractor shall select for test a one-gram sample from that quantity of pellets used in eight hours production of fuzes. If the volatile contents of the sample exceeds the requirement, the boron potassium nitrate pellets represented by the sample shall be rejected. The test shall be performed as specified in 4.3.2.1.

4.2.3.3 Volatile content of delay composition (Tungsten), (see 3.4.1), Major defect, Code No. 16001.-The contractor shall provide adequare controls to insure that all fuze powder used in the assembly comply with the volatile content requirements. For verification, the contractor shall select one sample of each sublot of powder used in each eight-hour loading production of fuze assemblies. A composite sample shall not be used. If the volatile content of the sample exceeds the requirement, and loading has begun, that quantity of the material represented by the sample shall be rejected. If fuze assemblies have been loaded with material containing excessive volatile content, the sample, together with all assemblies, shall be rejected. The test shall be performed as specified in 4.3.2.2.

4.2.3.4 Static Burning Time (see 3.6) - Major defect, Code No. 17001.-A sample of eighty (80) fuze assemblies (see dwg. 9232784) shall be selected at random from each fuze lot and tested to determine conformance to the requirements specified in 3.6. Forty (40) assemblies shall be tested at the 20 second setting and forty (40) assemblies tested at the 30 second setting. The lot shall be rejected if a total of two or more duds occur for both settings, or if the percent defective for burning times for each setting as computed with the use of MIL-STD-414, Section B, Table B3, AQL of 2.5 percent exceeds the maximum allowable percent defective. The tests shall be performed as specified in 4.3.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

4.2.3.5 Jumble (see 3.5), Critical defect, Code No. 18001.-Twelve sample fuze assemblies shall be selected from each lot for test as specified in 4.3.4. If any fuze assembly fails to comply with the requirements, the lot shall be rejected.

4.2.3.6 Functioning (proving ground)

4.2.3.6.1 Beginning with the first lot produced, and continuing until three consecutive lots have complied with the acceptance criteria, sixty (60) sample fuze assemblies shall be randomly selected and tested from each lot. Thirty (30) fuzes shall be tested at the 20 second setting and thirty (30) fuzes shall be tested at the 30 second setting. The lot shall be rejected if a total of three or more duds occur for both settings or if the percent defective for burning time as computed with the use of MIL-STD-414, Section B, Table B3, AQL 4.0 percent, exceeds the maximum allowable percent defective for each setting. The test shall be performed as specified in paragraph 4.3.5. The lot shall also be rejected if a critical defect occurs.

TABLE III

Defect	Classification	Code No.
Duds or no fuze action (see 3.7.1)	Major	11009
Premature Burst (see 3.7.2)	Critical	11010
Individual Burning time (see 3.7.3)	Major	11011

4.2.3.6.2 After three consecutive lots have complied with the criteria of 4.2.3.6.1, fifty (50) sample fuze assemblies shall be randomly selected and tested for all successive lots. Twenty-five, (25) fuzes shall be tested at the 20 second setting and twenty-five (25) fuzes shall be tested at the 30 second setting. The lot shall be rejected if a total of three or more duds occur for both settings or if the percent defective for burning time as computed with the use of MIL-STD-414, Section B, Table B3, AQL 4.0 percent, exceeds the maximum allowable percent defective for each setting. The test shall be performed as specified in paragraph 4.3.5. The lot shall also be rejected if a critical defect occurs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

4.2.3.7 Check test for deterioration of percussion primers.-If the total elapsed time between original acceptance of the percussion primer lot and the assembly of that lot into the fuze assemblies exceeds two years, or if the percussion primers have been subjected to adverse conditions for any period of time since previous test, the percussion primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-P-12951 (MU) immediately before the percussion primer lot is assembled into the fuze assemblies.

4.2.3.8 Torque on Adjustment Ring, Major defect, Code No. 11012 (see dwg. 9232784).-The torque at assembly necessary to turn the adjustment ring shall comply with the applicable requirement (75 to 90 inch pounds). Inspection shall be 100% using test equipment in accordance with 4.3.4.

4.2.3.9 Weight of Black Powder Expelling Charge (see dwg. 9232789) - Major defect - Code No. 08004.-The contractor shall provide adequate control to insure that the weight of the expelling charge added to the fuze body assembly complies with the requirements of drawing 9205597. The contractor shall select one fuze body every 15 minutes during loading at the expelling charge loading station to verify its conformance to requirements. If the weight of the block powder expelling charge in the sample selected does not meet the weight requirements, the fuze body assemblies represented by the 15 minute interval shall be rejected. The rejected fuze bodies shall be then check weighed and black powder shall be added or removed from the fuze body assemblies so that the expelling charge weight requirements are met.

4.2.4 Inspection equipment.-Index of Inspection Equipment List Number ET-9232784 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures

4.3.1 Moisture content

4.3.1.1 Black powder.-The moisture content of the black powder shall be determined in accordance with the procedures specified in Specification MIL-P-223.

4.3.2 Volatile content

4.3.2.1 Boron Potassium Nitrate Pellets.-The volatiles content of the boron potassium nitrate pellets shall be determined in accordance with Method 102.2 of MIL-STD-1234.

4.3.2.2 Delay composition.-The volatiles content of the tungsten delay composition shall be determined in accordance with Method 102.2 of MIL-STD-1234.

4.3.3 Static testing.-The fuze assemblies shall be conditioned at 75°F plus or minus 5°F, for a minimum of 4 hours, and tested at this temperature. All static tests will be conducted with Adjustment ring on fuze. The torque applied to the test head adapters to seat the fuze rings shall be such that the torque necessary to turn the adjustment ring shall be 75 inch pounds (min) to 90 inch pounds (max). The fuze assemblies shall be tested using a static burning time fuze tester which can be read directly to at least the nearest 0.005 seconds. All static tests shall be conducted with the expelling charge present in the fuze assembly. Each fuze tested shall be observed for the requirements specified in 3.6. Equipment shall be in accordance with 4.2.4.

4.3.4 Jumble.-The fuze assemblies shall be tested in accordance with the procedure specified in Standard MIL-STD-331, Test Method 102, except that the fuzes shall be assembled in a fixture as shown in Figure 1, prior to testing in the jumble boxes. Equipment used shall be in accordance with 4.2.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

4.3.5 Functioning (proving ground).-The fuze assemblies shall be assembled to an 81MM M301A3 Illuminating shell and fired from an 81MM Mortar at the quadrant elevation of 60° and charge zone 4 and 8. All flight times shall be determined by infra-red chronograph. The fuzes assembled to the cartridges shall be conditioned at 75° ± 5°F for a minimum of 4 hours prior to testing and fired at this temperature.

4.3.5.1 Test validity.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The fuze assemblies shall be preserved and packaged as specified on dwg. 8865235.

5.2 Packing

5.2.1 Level A.-The fiber container shall be packed as specified on dwg. 8865236.

5.3 Marking.-Marking shall be in accordance with dwgs. 8865235 and 8865236.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article sample.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET, in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

6.3.1 Inspection Equipment Lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock Numbers of existing equipment designs, or codes, as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment, as defined in 6.3.3 and 6.3.4, will be designated in the EL by the following codes:

.CD - Contractor design on controlled contractor format and/or commercial equipment
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified,

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

contractor designs may be developed on the format the contractor normally employs in his equipment design procedure, provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND1. All other designs of inspection equipment shall be approved by the inspection element of the agency admistering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissable and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design reveiw will be accomplished normally within one month after receipt.

6.4 Submission of test data.-In addition to the normal distribution of records, when the fuze is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to the following: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND2 and DJ-4, Dover, New Jersey, 07801.

6.5 Submission of first article sample.-Instructions as to the location for the evaluation of the first article samples shall be obtained from the contracting officer.

6.6 Statements of findings shall show the specific analytical results of each test or inspection performed as required by the applicable drawing or this specification, together with the specified acceptable limits.

Custodian:
ARMY-MU

Preparing Activity:
ARMY-MU

Project Number: 1315-A-639

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.

MIL-F-14716A(MU)
22 MARCH 1971

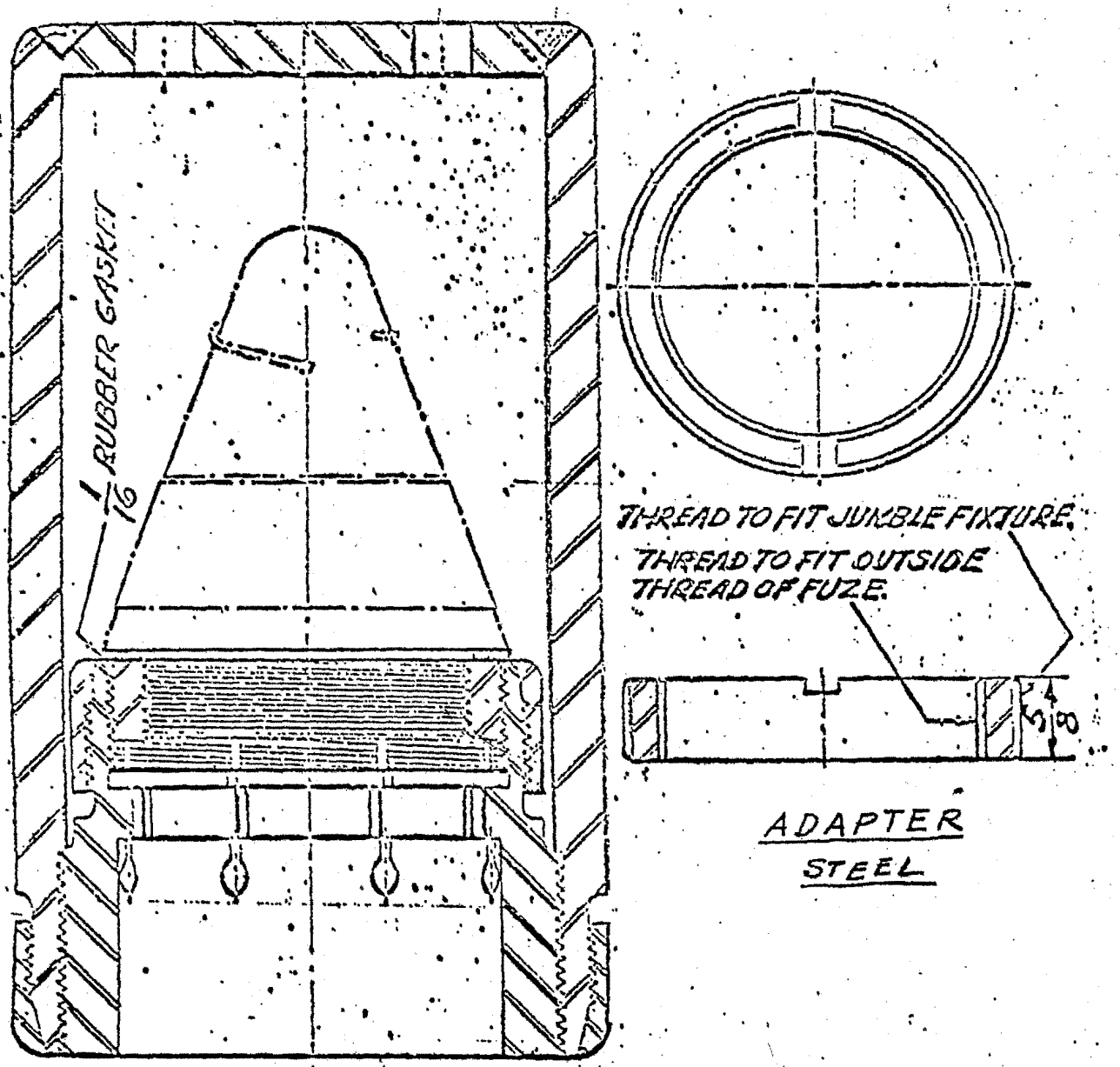


Fig 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:37Z
Check the source to verify that this is the current version before use.